

TRIAL H.S.C. EXAMINATION, AUGUST, 1998.

Mathematics - 3/4 Unit Common Paper

Name: _____ Class 12M
Student Number: _____

Time allowed:- 2 hours (+5 minutes reading time)

Instructions

- * All questions may be attempted.
- * All questions are of equal value, except Question 2 (15 marks).
- * All necessary working should be shown. Marks may not be awarded for careless or badly arranged work.
- * Approved calculators may be used.
- * Write your name, class and student number on this page, and your student number on each answer page.
- * Hand in this Question Paper at the end of the examination together with your answer pages in 3 bundles:
Bundle A - Questions 1,2,3;
Bundle B - Questions 4,5;
Bundle C - Questions 6,7.
- * This paper constitutes 40% of the school assessment, but does not necessarily reflect the format and content of the H.S.C.

Table of Standard Integrals

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \left\{ \frac{x}{a} \right\}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \left\{ \frac{x}{a} \right\}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln (x + \sqrt{x^2 - a^2}), \quad |x| > |a|$$

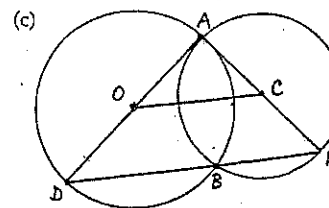
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln (x + \sqrt{x^2 + a^2})$$

Question 1 (14 marks)

- (a) The polynomial $P(x) = x^3 + 4x^2 + x + k$. When $P(x)$ is divided by $(x + 2)$, the remainder is 5. Find the value of k . [2]
- (b) Differentiate $\ln [3 \sin x]$. [3]
- (c) If $\log_A B = 2$, find $\log_B A^3$. [3]
- (d) Find the perpendicular distance between the lines $x - y + 3 = 0$ and $x - y + 1 = 0$. [3]
- (e) Using the substitution $u = \ln x$, find $\int \frac{1}{x\sqrt{4 - (\ln x)^2}} dx$. [3]

Question 2 (15 marks)

- (a) The function $f(\theta) = \cos \theta - \theta$ has a root close to 1. [2]
- (i) Show that the root lies between 0.7 and 0.8.
- (ii) Taking $\theta = 0.7$ as an approximation for the solution to the equation $\cos \theta = \theta$, use one application of Newton's method to give a better approximation. [3]
- (b) Show that $\int_0^3 \frac{dx}{9 + x^2} = \frac{\pi}{12}$. [3]



Two circles, centres O and C, intersect at A and B. Diameter ECA is a tangent to the circle with centre O. AOD is a diameter.

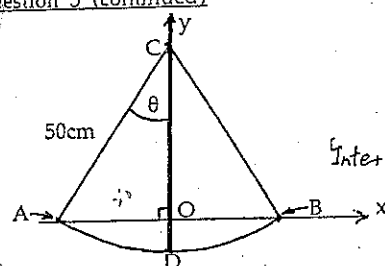
- (i) Show that D, B and E are collinear. [2]
- (ii) Prove that triangle ADB is similar to triangle EAB. [3]
- (iii) Show that $OC = \frac{1}{2} DE$. [2]

Question 3 (14 marks)

- (a) The day before a test, the probability that Student A is absent is 0.7. The probability that Student B is absent is 0.2 and the probability that Student C is absent is 0.4. Find the probability that, on a day before a test, Student A is present but Students B and C are both absent. [2]
- (b) Given that $\sin^{-1}[\cos x] = \theta$, where x, θ are acute, show that $\sin^{-1}[\cos x] = \cos^{-1}[\sin x]$. [4]

Question 3 (continued)

(c)



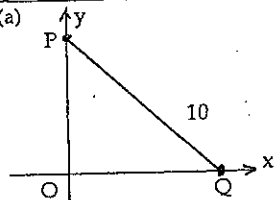
CA is a pendulum of length 50 cm oscillating uniformly about the vertical position, CD, so that the end of the pendulum describes the arc ADB and return in 4 seconds.

Interval AB = 60 cm and angle ACD = θ .

- Calculate the angle θ in radians. [1]
- Derive an equation to describe the horizontal motion of the end of the pendulum between A and B, starting at position A. [4]
- Calculate the area of the segment AOBD correct to two decimal places. [3]

Question 4 (14 marks)

(a)



PQ is a rod of fixed length 10 metres. Point P moves freely along the y-axis and point Q moves freely along the x-axis.

- Show that $\frac{dy}{dx} = -\frac{x}{\sqrt{100-x^2}}$. [2]
- Given that Q moves with constant velocity $\frac{dx}{dt} = 2 \text{ m/s}$, determine the velocity of P when $x = 5$ metres. [4]

(b)(i) The function $f(x) = \cos x e^{\sin x}$. Complete this table of values for $y = f(x)$.

x	0	0.5	1	1.5	0.5π	2	2.5	3	π
$f(x)$									

Hence draw a sketch of $y = f(x)$ for $0 \leq x \leq \pi$. [4]

- Calculate the area enclosed between the curve $y = \cos x e^{\sin x}$ and the x-axis from $x = 0$ to $x = \pi$. [4]

Question 5 (14 marks)

(a) In the time just before an election, the level of confidence in Candidate X is C, where C is expressed as a percentage. The rate of daily change in that confidence is given by $\frac{dC}{dt} = k(C + 0.1)$, where k is a constant.

- If C is initially 90%, show that $C = -0.1 + e^{kt}$. [3]
- If C = 60% 10 days later, find the value of k. [1]
- Determine the level of confidence in X at 20 days. [2]
- How many days will it take for the level of confidence in X to reduce to 10%? [2]

(b)(i) Sketch the curve $y = \cos^{-1} \left[\frac{x-2}{2} \right]$. [2]

- Show that when this curve is rotated about the y-axis, the volume generated is $6\pi^2$ units³. [4]

Question 6 (14 marks)

(a) \$A is borrowed at 12% per annum reducible interest, calculated monthly. The loan is repaid in equal monthly instalments of \$2400 over n months.

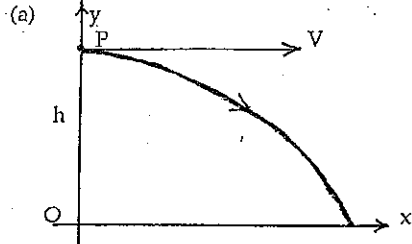
- Show that $A \times 1.01^n = 240000(1.01^n - 1)$. [3]
- If $A = 200000$, calculate the period of the loan correct to the nearest month. [3]
- Calculate the equivalent flat rate of interest per annum on the loan for the time calculated in part (ii). [2]

(b)(i) Show that $\frac{d}{d\theta} [\sec \theta + \tan \theta] = \sec \theta \tan \theta + \sec^2 \theta$. [2]

(ii) Using this result, show that $\int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + c$. [2]

(iii) Using the substitution $x = \sec \theta$, evaluate $\int_2^3 \frac{dx}{\sqrt{x^2-1}}$ correct to two decimal places. [2]

Question 7 (14 marks)



A particle is projected horizontally from a point P, h metres above O, with a velocity of V metres/second. The equations of motion of the particle are

$$\ddot{x} = 0 \text{ and } \ddot{y} = -10.$$

- (i) Show that the position of the particle at time t seconds is given by
 $x = Vt$ and $y = -5t^2 + h$.

A plane flying at a constant speed of 252 km/hr at a height of 145 metres above a horizontal stretch of land drops a package of supplies to a farm stranded by floods.

- (ii) How far will the package travel horizontally before hitting the ground? [3]
 (iii) If the package must clear a line of trees 20 metres high at the perimeter of the farm, what is the maximum horizontal distance the plane must be from the trees when it drops the package? [3]

- (b)(i) If $f(x) = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n + \dots$, where $0 < |x| < 1$, find a simpler expression for $f(x)$. [1]

- (ii) Find $\int_0^u f(x) dx$ and derive a series expansion for $\ln(1+u)$, where $0 < u < 1$. [3]

- (iii) Use the first four terms of this expansion to find an approximation for $\ln 1.6$. Do not use the calculator value of $\ln 1.6$, as your error will be detected. [2]

της ενδ όφ τησ εξαμιναιον εδεν ιφ ιτ ισ νοτ της ενδ όφ ψου αδιου

1)(a) $P(x) = x^3 + 4x^2 + x + k$
 $P(-2) = -8 + 16 - 2 + k = 5$
 $6 + k = 5$
 $k = -1$ [2]

(b) $\frac{d}{dx} \ln[3 \sin x] = \frac{3 \cos x}{3 \sin x}$
 $= \cot x$ [3]

(c) $\log_4 8 = 2$, so $B = A^{1/2}$
 $A = B^2$
 $\therefore \log_8 B = \log_8 B^{1/2}$ [14]
 $= \frac{1}{2}$ or 1.5 [3]

(d) On $x - y + 1 = 0$, $x = 0 \rightarrow y = 1$
 Distance from $(0, 1)$ to $x - y + 3 = 0$
 is $d = \frac{|0 - 1 + 3|}{\sqrt{1^2 + 1^2}} = \frac{2}{\sqrt{2}}$
 or $\sqrt{2}$ [3]

(e) $I = \int \frac{dx}{x \sqrt{4 - (\ln x)^2}}$, $u = \ln x$, $du = \frac{dx}{x}$
 $\therefore I = \int \frac{du}{\sqrt{4 - u^2}} + C$, C const.
 $= \sin^{-1}(\frac{u}{2}) + C$
 $= \sin^{-1}(\frac{\ln x}{2}) + C$ [3]

2)(a) $f(\theta) = \cos \theta - \theta$
 (i) $f(0.7) = 0.0648 \dots$
 $f(0.8) = -0.1033 \dots$ [2]
 $\therefore f(\theta)$ changes sign between 0.7 and 0.8.
 $\therefore f(\theta) = 0$ for $0.7 < \theta < 0.8$.

(ii) $\theta_2 = \theta_1 - \frac{f(\theta_1)}{f'(\theta_1)}$ [3]
 $= 0.7 - \frac{\cos 0.7 - 0.7}{-\sin 0.7 - 1}$
 $= 0.7 - \frac{0.0648 + 0.7}{-1.6422}$
 $= 0.7394$

(b) $\int_0^3 \frac{dx}{9 + x^2} = \int_0^3 \frac{dx}{3^2 + x^2}$
 $= [\frac{1}{3} \tan^{-1} \frac{x}{3}]_0^3$
 $= [\tan^{-1} 1 - \tan^{-1} 0] \cdot \frac{1}{3}$
 $= \frac{\pi}{12}$

(c)(i) $\angle ABD = 90^\circ$ (L in semicircle)
 $\angle ABE = 90^\circ$ (L in semicircle)
 $\therefore \angle DBE = 180^\circ$
 $\therefore DBE$ is a straight line. [2]
 $\therefore D, B, E$ are collinear.

(ii) In $\Delta ADB, EAB$
 $\angle ADB = \angle AEB = 90^\circ$ (proven)
 $\angle BAD = \angle EAB$ (L in semicircle)

$\therefore \Delta ADB, AEB$ are equiangular
 $\therefore \Delta ADB \parallel \Delta AEB$ (AAA sim.) [3]

(iii) In $\Delta AOC, ADE$,
 $\angle OAE$ is common
 $\frac{AO}{AD} = \frac{AC}{AE} = \frac{1}{2}$ [3]
 (radii = $\frac{1}{2}$ diameters)
 $\therefore \Delta AOC \parallel \Delta ADE$ (SAS sim.)
 $\therefore \frac{OC}{DE} = \frac{1}{2}$ (sides in prop)
 $\therefore OC = \frac{1}{2} DE$

3)(a) $P(A, B, C) = 0.3 \times 0.2 \times 0.4$
 $= 0.024$ [2]

(b) $\sin^{-1}[\cos x] = \theta$
 $\therefore \sin \theta = \cos x$
 $\therefore \theta = \frac{\pi}{2} - x$
 $\therefore \cos \theta = \sin x$
 ie. $\cos^{-1}[\sin x] = \theta$ [4]
 Hence $\sin^{-1}[\cos x] = \cos^{-1}[\sin x]$.

(c) ΔABC is a right-angled triangle with $\angle C = 90^\circ$
 $AC = 30 \text{ cm}$, $BC = 30 \text{ cm}$
 $\therefore \sin \theta = \frac{30}{50} = 0.6$
 $\therefore \theta = 0.6435$

(ii) Since motion is simple harmonic, equation is of form $x = a \cos(\omega t + \epsilon)$.
 Amplitude, $a = 30$.
 Period $2\pi/\omega = 4$ [14]
 $\therefore \omega = \frac{\pi}{2}$.

$\therefore x = 30 \cos(\frac{\pi}{2}t + \epsilon)$
 When $t = 0$, $x = -30$.
 $\therefore -30 = 30 \cos \epsilon$
 $\cos \epsilon = -1$
 $\epsilon = \pi$ [4]
 $\therefore x = 30 \cos(\frac{\pi}{2}t + \pi)$
 OR $x = -30 \cos \frac{\pi}{2}t$

Alternatively, using $x = a \sin(\omega t + \epsilon)$
 $x = 30 \sin(\frac{\pi}{2}t + \epsilon)$
 $-30 = 30 \sin \epsilon$
 $\sin \epsilon = -1$
 $\epsilon = -\frac{\pi}{2}$
 $\therefore x = 30 \sin \frac{\pi}{2}(t - 1)$

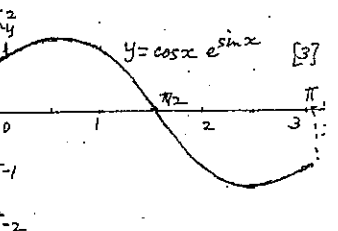
3e(iii) Segment $AB = \frac{1}{2} \times 20 = 10$
 Area $= \frac{1}{2} \times 50^2 \times (\frac{1}{2} \times 20 - \sin 1.22)$
 $= 108.75 \text{ cm}^2$ [6]

4(a) $x^2 + y^2 = 100$
 $\therefore y = \sqrt{100 - x^2}$
 $\therefore \frac{dy}{dx} = \frac{-x}{\sqrt{100 - x^2}}$
 $\therefore \frac{dy}{dx} = -\frac{x}{\sqrt{100 - x^2}}$ [6]

(ii) $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$
 $= -\frac{x}{\sqrt{100 - x^2}} \times 2$ [14]
 At $x = 5$, $\frac{dy}{dt} = -\frac{5}{\sqrt{100 - 25}} \times 2$
 $= -\frac{10}{5\sqrt{3}}$
 $= -\frac{2}{\sqrt{3}}$ or $-\frac{2\sqrt{3}}{3} \text{ m/s}$.

(b)(i) $f(x) = \cos x \cdot e^{\sin x}$ [17]

x	0	0.5	1	1.5	0.5\pi	2	2.5	3
f(x)	1	1.42	1.25	1.19	0	-1.03	-1.44	-1.44



(ii) Area $= \int_0^{\pi/2} \cos x e^{\sin x} dx + \int_{\pi/2}^{\pi} \cos x e^{\sin x} dx$
 $= [e^{\sin x}]_0^{\pi/2} + [e^{\sin x}]_{\pi/2}^{\pi}$
 $= (e - 1) + (1 - e)$
 $= 2e - 2 \text{ units}^2$
 $\approx 3.4366 \text{ units}^2$ [4]

5. (a)

$$C = -0.1 + e^{kt}$$

$$\frac{dC}{dt} = k e^{kt}$$

$$= k(-0.1 + e^{kt} + 0.1)$$

$$= k(C + 0.1)$$

When $t = 0$

$$C = -0.1 + 1$$

$$= 0.9$$

$$= 90\%$$

(ii) When $t = 10$; $C = 0.6$

$$0.6 = -0.1 + e^{10k}$$

$$\therefore k = \frac{1}{10} \ln 0.7$$

$$\div -0.035667494$$

(iii) When $t = 20$, $C = -0.1 + e^{20k}$

$$\therefore C = 0.39$$

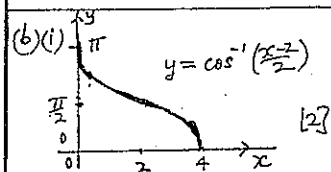
$\therefore$ Level of confidence is 39%

(iv) $C = 0.1$ $\therefore 0.1 = -0.1 + e^{kt}$

$$\therefore t = \frac{10 \ln 0.2}{\ln 0.7}$$

$$t = 45.123 \text{ days.}$$

$\therefore$ Level of confidence reaches 10% on the 46th day.



(ii) Volume = $\int_0^4 \pi x^2 dy$

Now $\frac{x-2}{2} = \cos y$

$$x = 2 \cos y + 2$$

$$\therefore V = \pi \int_0^{\pi/2} (4 \cos^2 y + 4 \cos y + 4) dy$$

$$V = \pi \int_0^{\pi/2} (2 \cos 2y + 4 \cos y + 6) dy$$

$$= \pi \left[\sin 2y + 4 \sin y + 6y \right]_0^{\pi/2}$$

$$= \pi \{ (0 + 0 + 6\pi) - (0 + 0 + 0) \}$$

$$= 6\pi^2 \text{ units}^3$$

6(a) Let A_n be amount owing at end of n months.

(i) $A_1 = 1.01A - 2400$

$$A_2 = 1.01A_1 - 2400$$

$$= 1.01^2 A - 2400(1.01 + 1)$$

etc.

$$A_n = 1.01^n A - 2400(1.01^{n-1} + \dots + 1.01 + 1)$$

$$A_n = 0$$

$$\therefore A \times 1.01^n = \frac{2400 \times (1.01^n - 1)}{1.01 - 1}$$

$$A \times 1.01^n = 240000(1.01^n - 1)$$

(ii) $A = 200000$

$$\therefore 200000 \times 1.01^n = 240000(1.01^n - 1)$$

$$\therefore 40000 \times 1.01^n = 240000$$

$$1.01^n = 6$$

$$n = \frac{\log 6}{\log 1.01}$$

$$= 180.07$$

$\therefore$ Period of loan is 180 months

or 15 years.

(iii) Interest = $\$2400 \times 180$

$$= \$232000$$

$$\text{Interest per } \% = \frac{232000}{15} \times \frac{100}{200000}$$

$$= 7.73\%$$

(b)(i) $\frac{d}{d\theta} [\sec \theta + \tan \theta] = \frac{1}{\cos^2 \theta} + \sec^2 \theta$

$$= -(\cos \theta)^{-2} (\sin \theta) + \sec^2 \theta$$

$$= \frac{\sin \theta}{\cos^3 \theta} + \sec^2 \theta$$

$$= \sec \theta \tan \theta + \sec^2 \theta$$

(ii) $\int \sec \theta d\theta = \int \frac{\sec \theta (\sec \theta + \tan \theta) d\theta}{(\sec \theta + \tan \theta)}$

$$= \int \frac{\sec^2 \theta + \sec \theta \tan \theta}{\sec \theta + \tan \theta} d\theta$$

$$= \ln |\sec \theta + \tan \theta| + C$$

where C is a constant.

(iii) $I = \int_2^3 \frac{dx}{\sqrt{x^2 - 1}}$

Let $x = \sec \theta$, so $dx = \sec \theta \tan \theta d\theta$

Then $I = \int_{\sec^{-1} 2}^{\sec^{-1} 3} \frac{\sec \theta \tan \theta d\theta}{\sqrt{\sec^2 \theta - 1}}$

$$= \int_{\sec^{-1} 2}^{\sec^{-1} 3} \frac{\sec \theta \tan \theta}{\tan \theta} d\theta$$

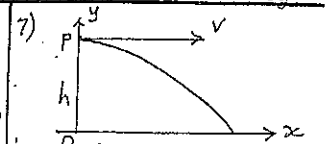
$$= \int_{\sec^{-1} 2}^{\sec^{-1} 3} \sec \theta d\theta$$

$$= [\ln (\sec \theta + \tan \theta)]_{\sec^{-1} 2}^{\sec^{-1} 3}$$

$$= \ln (3 + 2\sqrt{2}) - \ln (2 + \sqrt{3})$$

$$= 1.7627 - 1.3170$$

$$= 0.45 \text{ (to 2 dec.pl.)}$$



(i) $\ddot{x} = 0$ $\ddot{y} = -10$

$$\dot{x} = v$$

$$x = vt + c_1$$

$$t = 0 \rightarrow x = 0$$

$$\therefore c_1 = 0$$

$$\therefore x = vt$$

$$y = -5t^2 + c_2$$

$$t = 0 \rightarrow y = h$$

$$\therefore c_2 = h$$

$$\therefore y = -5t^2 + h$$

(ii) $V = 252 \text{ km/h} = 252 \div 3.6 = 70 \text{ m/s}$

$$h = 145 \text{ m}$$

At ground, $y = 0$

$$\therefore 0 = -5t^2 + 145$$

$$5t^2 = 145$$

$$t^2 = 29$$

$$t = 5.385 \text{ sec.}$$

$$\therefore x = 70 \times 5.385$$

$$= 376.96 \text{ m.}$$

$\therefore$ Package takes 376.96 m or approximately 377 m.

(iii) When $y = 20$, $20 = -5t^2 + 145$

$$5t^2 = 125$$

$$t = 5 \text{ sec.}$$

$$\therefore x = 70 \times 5$$

$$= 350 \text{ m.}$$

$\therefore$ Plane must be no more than 350 m horizontally from the trees.

(b)(i) $f(x) = 1 - x + x^2 - x^3 + \dots + (-1)^n x^n$

Since $|x| < 1$, this is a GP with a limiting sum.

$$\therefore f(x) = \frac{1}{1+x}$$

(ii) $\int_0^1 f(x) dx = \int_0^1 \frac{1}{1+x} dx$

$$= [\ln(1+x)]_0^1$$

$$= \ln(1+1)$$

$$= \ln 2$$

But $\int_0^1 f(x) dx = \int_0^1 (1 - x + x^2 - x^3 + \dots) dx$

$$= \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{(-1)^n x^{n+1}}{n+1} + \dots \right]_0^1$$

$$= 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^n}{n+1} + \dots$$

$$\therefore \ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^n}{n+1} + \dots$$

$$(iii) \ln 1.6 \div 0.6 - \frac{0.6^2}{2} + \frac{0.6^3}{3} - \frac{0.6^4}{4}$$

$$= 0.4596$$

Total = 100 marks